In [ ]:
```
import numpy as np
import matplotlib.pyplot as py
import pandas as pd
```

In [ ]:
```
dataset=pd.read_csv("Social_Network_Ads.csv")
```

In [ ]:
```
dataset
```

Out[7]:

	User ID	Gender	Age	EstimatedSalary	Purchased
0	15624510	Male	19	19000	0
1	15810944	Male	35	20000	0
2	15668575	Female	26	43000	0
3	15603246	Female	27	57000	0
4	15804002	Male	19	76000	0
...	...	...	...	...	...
395	15691863	Female	46	41000	1
396	15706071	Male	51	23000	1
397	15654296	Female	50	20000	1
398	15755018	Male	36	33000	0
399	15594041	Female	49	36000	1

400 rows × 5 columns

In [ ]:
```
dataset=pd.get_dummies(dataset,drop_first=True)
dataset
```

Out[8]:

	User ID	Age	EstimatedSalary	Purchased	Gender_Male
0	15624510	19	19000	0	True
1	15810944	35	20000	0	True
2	15668575	26	43000	0	False
3	15603246	27	57000	0	False
4	15804002	19	76000	0	True
...	...	...	...	...	...
395	15691863	46	41000	1	False
396	15706071	51	23000	1	True
397	15654296	50	20000	1	False
398	15755018	36	33000	0	True
399	15594041	49	36000	1	False

400 rows × 5 columns

In [ ]:
```
dataset = dataset.drop("User ID", axis=1)
dataset["Purchased"].value_counts()
```

Out[10]:

	count
Purchased	
0	257
1	143

dtype: int64

In [ ]:
```
indept=dataset[["Age","EstimatedSalary","Gender_Male"]]
dep=dataset["Purchased"]
```

In [ ]:
```
from sklearn.model_selection import train_test_split

# Split the dataset into 75% Training set and 25% Test set
x_train, x_test, y_train, y_test = train_test_split(indept, dep,
test_size=0.25, random_state=0)

print(f"Training set shape: {x_train.shape}")
print(f"Test set shape: {x_test.shape}")
```

```
Training set shape: (300, 3)
Test set shape: (100, 3)
```

In [ ]:
```
from sklearn.preprocessing import StandardScaler

sc = StandardScaler()
x_train = sc.fit_transform(x_train)
x_test = sc.transform(x_test)

print("Features scaled successfully. Here is a sample of x_train:")
print(x_train[:5])
```

```
Features scaled successfully. Here is a sample of x_train:
[[ 0.58164944 -0.88670699 -0.98019606]
 [-0.60673761  1.46173768  1.02020406]
 [-0.01254409 -0.5677824  -0.98019606]
 [-0.60673761  1.89663484 -0.98019606]
 [ 1.37390747 -1.40858358 -0.98019606]]
```

In [ ]:
```
from sklearn.metrics import classification_report
clf_report=classification_report(y_test, y_pred)
print(clf_report)
```

```
              precision    recall  f1-score   support

           0       0.90      0.96      0.93        68
           1       0.89      0.78      0.83        32

    accuracy                           0.90       100
   macro avg       0.90      0.87      0.88       100
weighted avg       0.90      0.90      0.90       100
```

In [ ]:
```
from sklearn.metrics import confusion_matrix
cm = confusion_matrix(y_test, y_pred)
print(cm)
```

```
[[65  3]
 [ 7 25]]
```

In [ ]:
```
classifier.predict([[40,300,1,]])
```

Out[25]:
```
array([1])
```